



A DUT Reference Clock Jitter Attenuator Module for the V93000 ATE Platform (E5214B)

UPDATED FOR THE E5214B rev. B

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Abstract

In this paper we will describe a module developed to be integrated on a standard V9300 DUT loadboard that is used to provide a very low jitter reference clock to a DUT using a low cost V93000 channel (e.g. PinScale 1600). The module is able to reduce the reference clock jitter from typical values of 3 to 4 ps RMS for a low speed ATE channel to below 400 fs. The module only requires two connectors with a specific footprint to be integrated on the DUT loadboard. A total of nine low-speed channels including the reference clock are needed by the jitter attenuator module to generate a low phase noise reference clock plus 5 V from the utility power and a utility control bit. The jitter attenuator module also provides DC-Access for parametric measurements.

Key Words- DUT Reference clock, Jitter Attenuator, Low Phase Noise, E5214B.

1. Introduction

Not only, but specifically for devices that contain high-speed IO (HSIO) interfaces, the performance of the DUT's reference clock, especially its jitter content, is critical for proper testing of the DUT. This is the case in particular for characterization measurements. However, it also applies to loopback-based production testing in order to achieve the specified HSIO transmitter performance since jitter content of the reference clock is transferred to the timing of the DUT's transmitters. Thus, for several HSIO test implementations the reference clock often is provided by high-speed low jitter ATE channels like Pin Scale SL although the frequency requirements of the reference clocks typically are only in the few 100 MHz range. The use of high-speed ATE channels for this purpose often is considered overkill in light of the fact that multi-Gbit/s channels are used for such low frequency signal generation. There are previous proposals to use low-cost lower data rate channels in combination with e.g. crystal oscillators that help to reduce the native jitter of these channels. These proposals however have the drawback that they are limited to one dedicated reference clock frequency. For HSIO testing where multi-purpose PHY-IPs are used to cover a range of HSIO standards, the reference clock frequencies of one DUT however are not necessarily limited to one specific frequency. For such kind of devices, the reference clock frequencies can vary also for the same DUT depending on the final application the device is targeted for.

In order to address this requirement, we propose in this paper a solution to provide low jitter reference clocks from low-cost lower speed ATE channels like PinScale 1600 (PS1600) that offer variable frequency settings for the DUT reference clock. We use a jitter attenuator IC that is PLL-based (Silicon Labs Si5326). The circuit was implemented on a small daughter board or module

that easily can be plugged on any DUT loadboard that has two daughter board connectors with appropriate footprint designed in. Figure 1 shows a picture of the module rev A which is now obsolete. Figure 2 shows the new revision (rev. B) of the module. This new revision uses a lower profile relay to reduce the module height so that it can be used without problems in a direct dock wafer probing setup. There is no performance difference between the rev. A and rev. B modules but the loadboard connector is different (see Section 2).

Figure 3 shows a high-level block diagram of the jitter attenuator module.

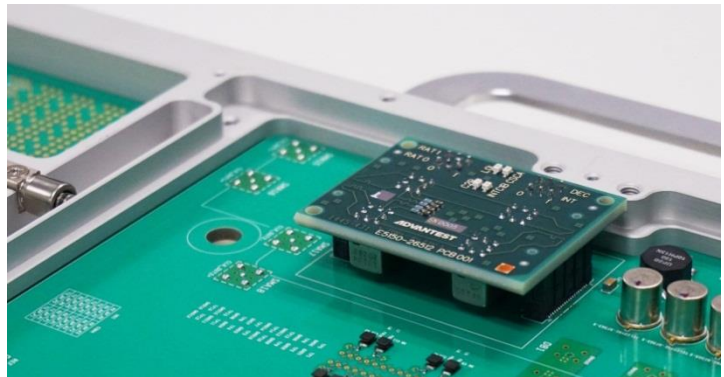
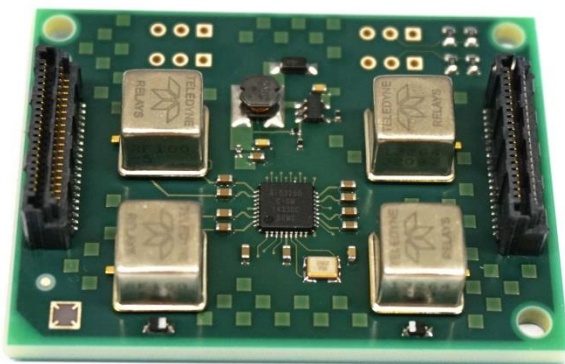


Figure 1: Jitter attenuator module rev. A (left) and integrated on the bottom side of a DUT loadboard (right). This module version is obsolete due to its height.



Figure 2: Jitter attenuator module rev. B. This new module version was designed using a lower profile relay to reduce its overall height. Advantest product number E5214B

As shown in the Figure 3 block diagram, the jitter attenuator module offers a bypass mode to allow DC tests on the DUT reference clock input pins. The connections to the board are kept to a minimum. Besides the clock input and output signals, the only other signals are a SPI interface that is used to configure the jitter attenuator via the V93000 protocol aware SPI implementation.

The jitter attenuator module core is a Silicon Labs Si5326 jitter attenuator device. One important point is that the output amplitude of this device is fixed although there are several options as shown in Figure 4. The test engineer needs to choose the option that is most appropriate to his application. The details on how to setup the right output levels will be presented later in this paper.

In the next section we will cover the DUT loadboard footprint requirements in detail followed by some PCB adapters for the jitter attenuator module that can be very useful. We will then discuss the concept we apply to determine optimal jitter attenuator settings and how these settings are transferred conveniently to the module. We will then conclude this paper with several performance results for the jitter attenuator module in relation to all digital Smart Scale

cards and the PortScale low phase noise source as well as commercially available low phase noise clock solutions designed for bench characterization.

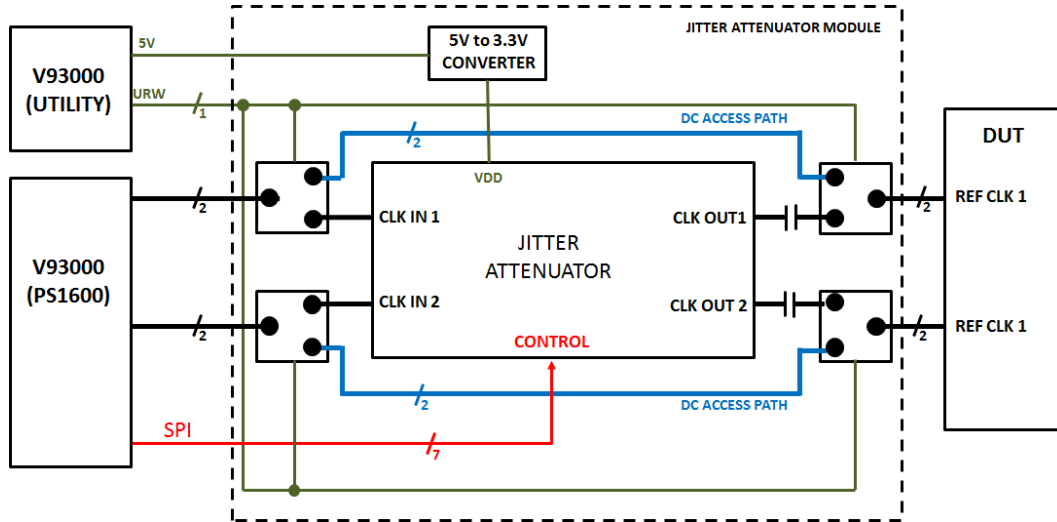


Figure 3: Jitter attenuator module high level block diagram.

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Differential Output Swing	CKO_{VD}	LVPECL 100 Ω load line-to-line	1.1	—	1.9	V_{PP}
Single Ended Output Swing	CKO_{VSE}	LVPECL 100 Ω load line-to-line	0.5	—	0.93	V_{PP}
Differential Output Voltage	CKO_{VD}	CML 100 Ω load line-to-line	350	425	500	mV_{PP}
Common Mode Output Voltage	CKO_{VCM}	CML 100 Ω load line-to-line	—	$V_{DD}-0.36$	—	V
Differential Output Voltage	CKO_{VD}	LVDS 100 Ω load line-to-line	500	700	900	mV_{PP}
		Low Swing LVDS 100 Ω load line-to-line	350	425	500	mV_{PP}
Common Mode Output Voltage	CKO_{VCM}	LVDS 100 Ω load line-to-line	1.125	1.2	1.275	V
Differential Output Resistance	CKO_{RD}	CML, LVPECL, LVDS	—	200	—	Ω
Output Voltage Low	CKO_{VOLLH}	CMOS	—	—	0.4	V
Output Voltage High	CKO_{VOHLH}	$V_{DD} = 1.71$ V CMOS	$0.8 \times V_{DD}$	—	—	V
Notes: - Current draw is independent of supply voltage - No under- or overshoot is allowed. - LVPECL outputs require nominal $V_{DD} \geq 2.5$ V. - This is the amount of leakage that the 3-Level inputs can tolerate from an external driver. See Si53xx Family Reference Manual for more details. - LVPECL, CML, LVDS and low-swing LVDS measured with $F_o = 622.08$ MHz.						

Figure 4: Silicon Labs Si5326 DUT reference clock output amplitude options (V_{DD} in the jitter attenuator module is 3.3 V).

2. DUT Loadboard Pin Assignment and Mechanical Footprint Requirements

To use the jitter attenuator module (Advantest product number E5214B) it is necessary to design the appropriate docking footprint on the DUT loadboard with the corresponding pin assignment. Figure 5 shows the required footprint and Figure 6 the pin assignment for each connector. The connector that needs to be integrated on the DUT loadboard for the jitter module is a Samtec LSEM-120-03.0-F-DV-A-N-K-TR. A DXF file version of the footprint is available on the V93000 online documentation. Note that for the obsoleted rev. A of the module a Samtec LSEM-120-06.0-F-DV-A-N-K-TR connector is needed on the loadboard (6 mm instead of 3 mm with rev. B). This means that rev. B module can be attached to loadboards designed for rev A. modules but the older rev. A modules cannot be attached to loadboards using the new 3 mm connector.

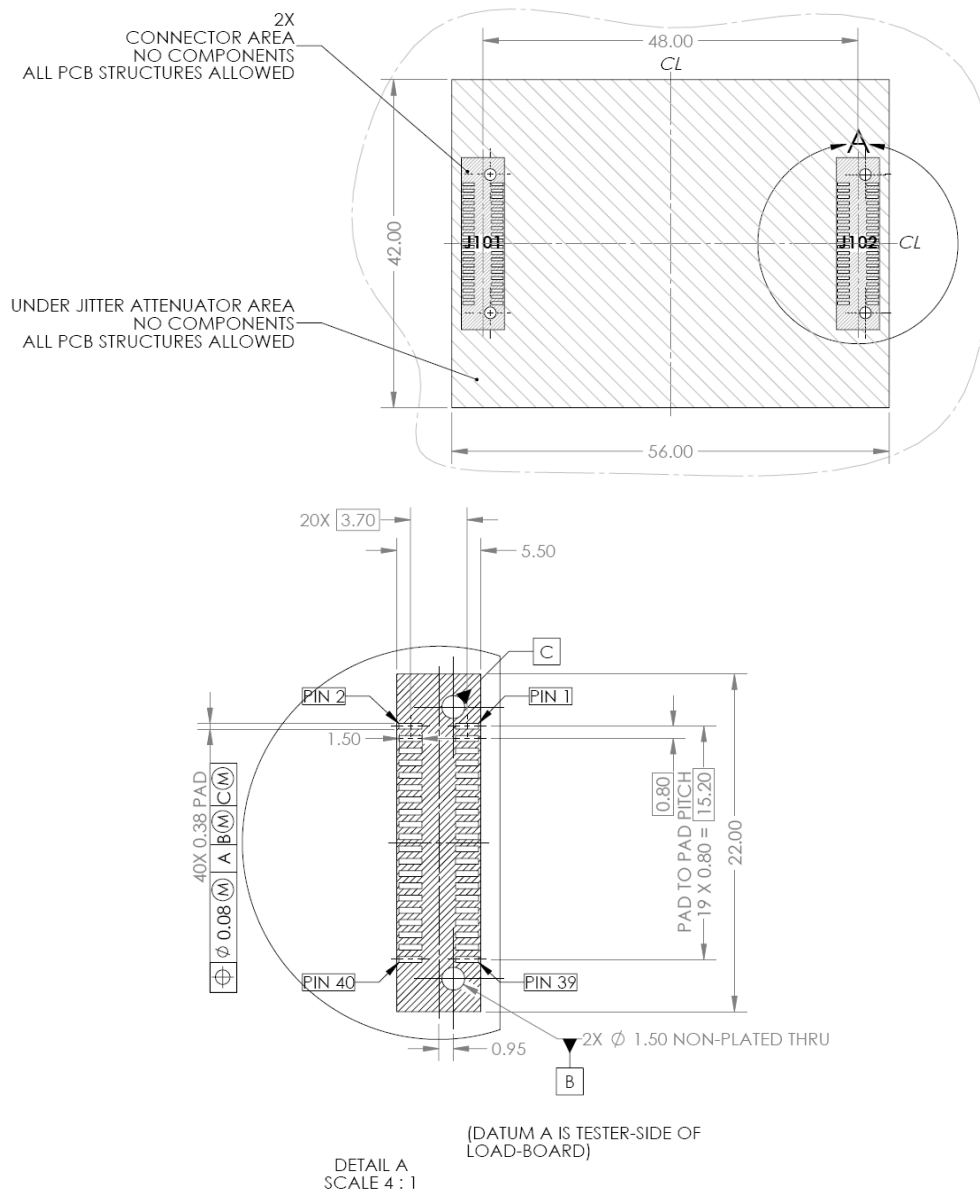


Figure 5: DUT loadboard footprint requirement for the jitter attenuator module (E5214B).

J101		J102	
PIN#	NAME	PIN#	NAME
1	5V	1	GND
2	5V	2	GND
3	GND	3	GND
4	REL_CTRL	4	GND
5	GND	5	GND
6	A2_SS	6	GND
7	GND	7	GND
8	A0	8	GND
9	GND	9	GND
10	SCL	10	GND
11	GND	11	GND
12	SDO	12	GND
13	GND	13	CLK_OUT1-
14	A1	14	GND
15	GND	15	CLK_OUT1+
16	SDI	16	GND
17	GND	17	GND
18	NRST	18	GND
19	GND	19	GND
20	GND	20	GND
21	GND	21	GND
22	GND	22	GND
23	GND	23	CLK_OUT2-
24	CLK_IN1-	24	GND
25	GND	25	CLK_OUT2+
26	CLK_IN1+	26	GND
27	GND	27	GND
28	GND	28	GND
29	GND	29	GND
30	GND	30	GND
31	GND	31	GND
32	GND	32	GND
33	GND	33	GND
34	CLK_IN2-	34	GND
35	GND	35	GND
36	CLK_IN2+	36	GND
37	GND	37	GND
38	GND	38	GND
39	GND	39	GND
40	GND	40	GND

Figure 6: Jitter attenuator module pin assignment.

2. UDI Jitter Attenuator Adapter (E5150-66424)

A special adapter was developed to allow the easy use of the jitter attenuator module in the context of the Universal DUT interface (UDI) framework [2]. Figure 7 shows a picture of the UDI jitter attenuator adapter module (Advantest internal part number E5150-66424) and also the adapter with the jitter attenuator module installed.

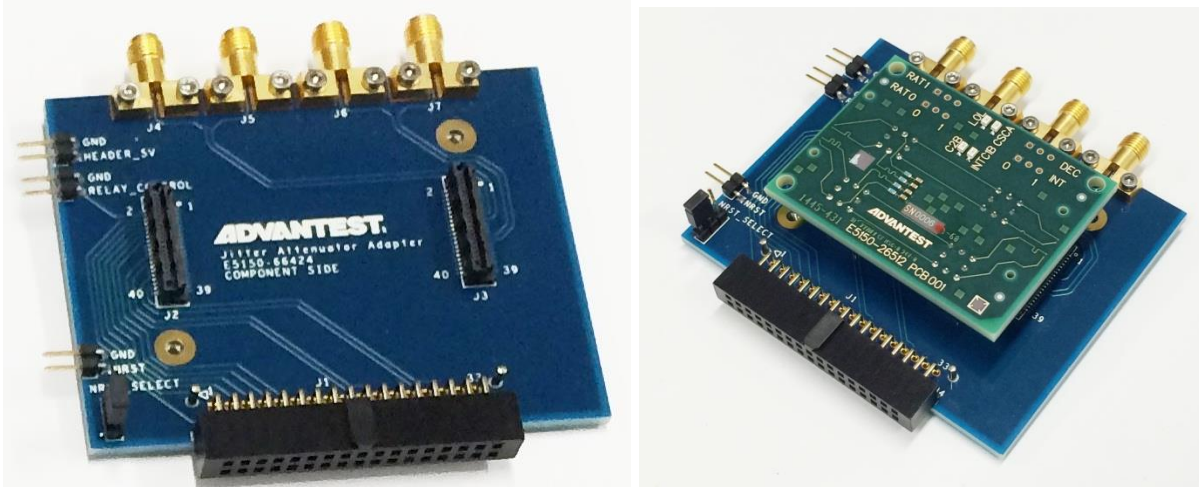


Figure 7: Jitter attenuator UDI adapter (left) and with the jitter attenuator module attached (right)

The jitter attenuator adapter was designed to be installed in a V93000 hybrid stiffener (N2360-63014/63015) and a male header anti-pogo PCB (E5150-26400) as shown in Figure 8. This setup allows the jitter attenuator UDI adapter to be installed for any pogo location. A utility/MS-DPS UAB2 (E5150-66430) is also needed to provide the 5 V power supply to the jitter attenuator module and also to control the bypass path relays. Table 1 provides a description of the jitter attenuator pin assignment when using the jitter attenuator module UDI adapter.



Figure 8: Jitter attenuator UDI adapter with attached jitter attenuator module being used with a UDI hybrid stiffener for measuring the jitter attenuator module performance. Notice the balun used on the setup to convert the differential output from the jitter attenuator modules to a single-ended signal for the measurement instrument.

Table 1: Jitter Attenuator UDI Adapter Pin Assignment.

Channel Number	Name	Notes
01	A2_SS	
02	A0	
03	SCL	
04	SD0	
05	A1	
06	SDI	
07	nRST	Optional
09	CLK_IN1-	
11	CLK_IN1+	
13	CLK_IN2-	
15	CLK_IN2+	

The two differential clock outputs are routed to SMA connectors on the top of the UDI adapter PCB. The test engineer can then easily connect bench instrument like a spectrum analyzer or an oscilloscope to it. This setup is ideal for easy verification and demonstration of the jitter attenuator performance for a give reference clock setup. The other possible use of the jitter attenuator UDI adapter is inside an UDI with a V93000 top interface [4]. In this case a pogo pin to SMA assembly is used to connect the jitter attenuator UDI adapter outputs to pogo pin location on top of the UDI V93000 top interface.

The jitter attenuator UDI adapter was designed to be used with the V93000. For measuring the jitter attenuator module in a bench instrumentation setting, a jitter attenuator motherboard (E5150-26513) was developed that allows access to all jitter attenuator pins thru SMA and male header connectors. Figure 9 shows a picture of the jitter attenuator motherboard.

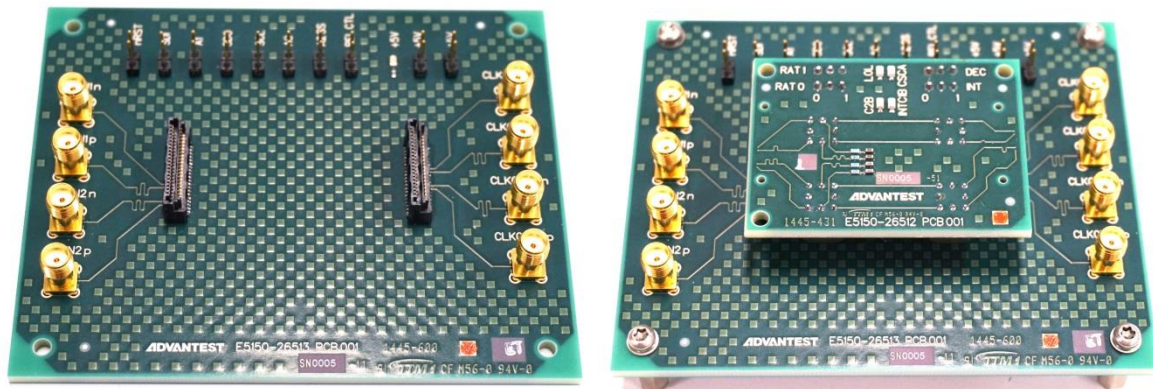


Figure 9: Jitter attenuator motherboard for bench usage.

3. Jitter Attenuator Thru Adapter (E5150-66431)

Another module developed in the context of the jitter attenuator module is the jitter attenuator thru adapter (Advantest internal part number E5150-66431). The objective of this module is to provide an AC-coupled thru connection between the V93000 clock inputs and the jitter attenuator clock outputs. Figure 10 shows a picture of the jitter attenuator thru adapter. The thru lines are AC coupled with a 10 nF capacitor.

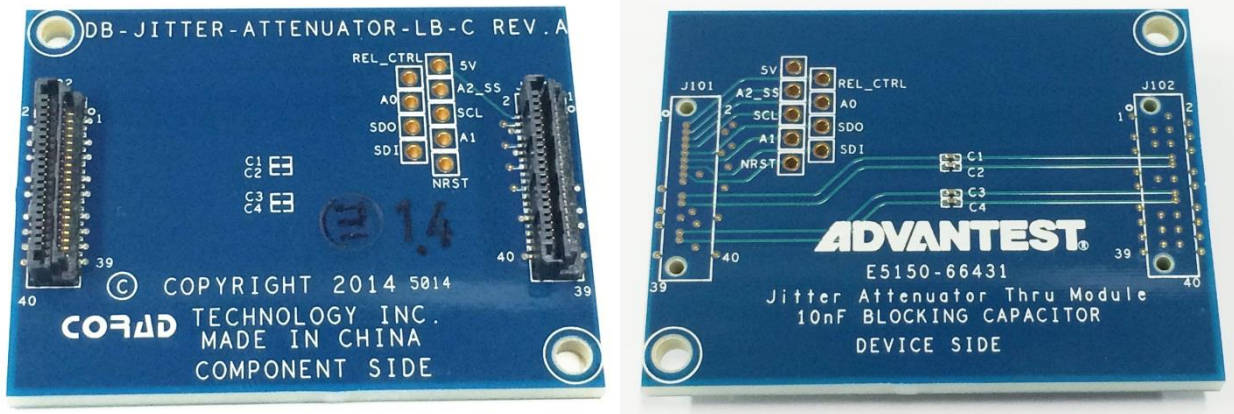


Figure 10: Jitter attenuator thru adapter.

Typical usage examples of the jitter attenuator thru adapter are checking the DUT loadboard jitter attenuator footprint is OK or for debug reasons using it instead of the jitter attenuator module.

4. Programming the PLL Parameters and Smartest SW Integration

The jitter attenuator module PLL is programmed via an SPI interface using the V9300 Smartest protocol editor. The jitter attenuator IC manufacturer Silicon Labs provides a PC based wizard program that calculates appropriate PLL divider settings for the desired input/output clock frequency settings. In addition, this wizard allows export of an ASCII device register map file with the required register settings for the desired device settings. The program can be downloaded using the link [5]. After installation the DSPLLsim program can be executed resulting in the initial window shown in Figure 1. The user should select to create a new frequency plan.

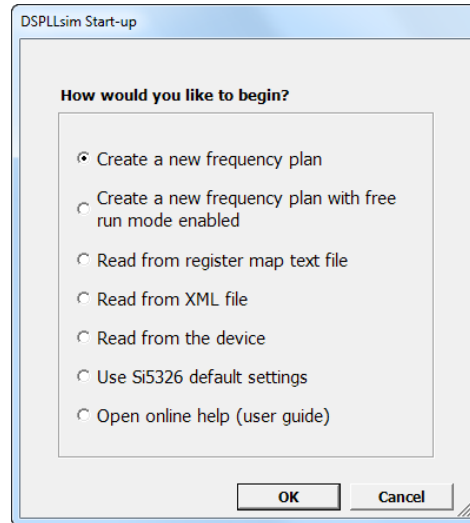


Figure 11: Initial window from the DSPLLsim program.

After pressing OK the window in Figure 12 appears where the user needs to select the correct device which is the Si5326.

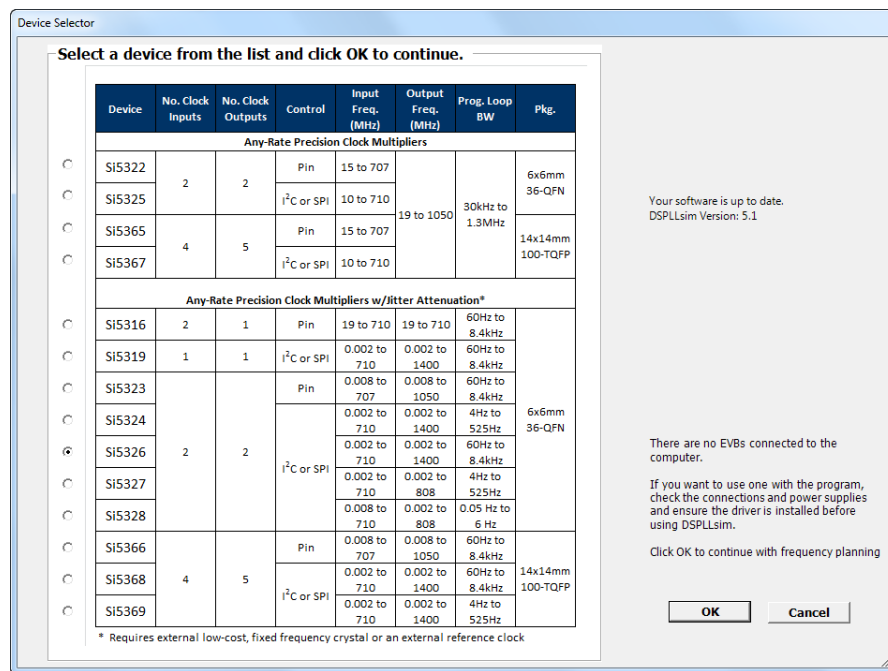


Figure 12: Selecting the correct device.

After pressing OK the window in Figure 13 appears where the user needs to select the input frequency from the V93000 channel and the intended output frequency. Although usually this would be a one to one ratio for most applications with a single reference clock, it is also possible to use the jitter attenuator module to provide two reference clocks with different frequencies that are not matching the input frequency.

DSPLLsim: Frequency Planner for Si5326

Configuration

Enter the multiplication factor as a ratio of two integers and then press the button underneath it to calculate the output frequency; or enter the desired output frequency and press the button underneath it to calculate the multiplication ratio. In both cases, be aware that the ratio is always exact; but that the input and output frequencies are rounded to the nearest Hz.

CKIN1 (MHz) X $\frac{\text{CKOUT1 to CKIN1 Ratio}}{\frac{1}{1}}$ = CKOUT1 (MHz)

Specify the number of input clocks.

Specify the number of output clocks.

☐ Force inexact mode. This calculates output clock frequency error in ppm for frequency plans in which an exact solution is not available.

Figure 13: Selecting the input and output clock frequencies and number of input/output clocks.

If two input clock frequencies are selected or two output clocks are selected, it is then necessary to also define the ratio of those frequencies to the CLKIN1 frequency as shown in Figure 14. Finally after pressing next the frequency and divider setting results are shown in a window as shown in the example of Figure 15.

DSPLLsim: Frequency Planner for Si5326

Input Clocks

Enter the ratio for each input clock.

CKIN1 (MHz) X $\frac{\text{CKIN2 to CKIN1 Ratio}}{\frac{1}{1}}$ = CKIN2 (MHz)

Note: These are ratios of integers and are exact.

DSPLLsim: Frequency Planner for Si5326

Output Clocks

Enter the ratio for each output clock.

CKOUT1 (MHz) X $\frac{\text{CKOUT2 to CKOUT1 Ratio}}{\frac{1}{1}}$ = CKOUT2 (MHz)

Note: These are ratios of integers and are exact.

Figure 14: Settings for the second input reference clock (left) and output reference clock (right).

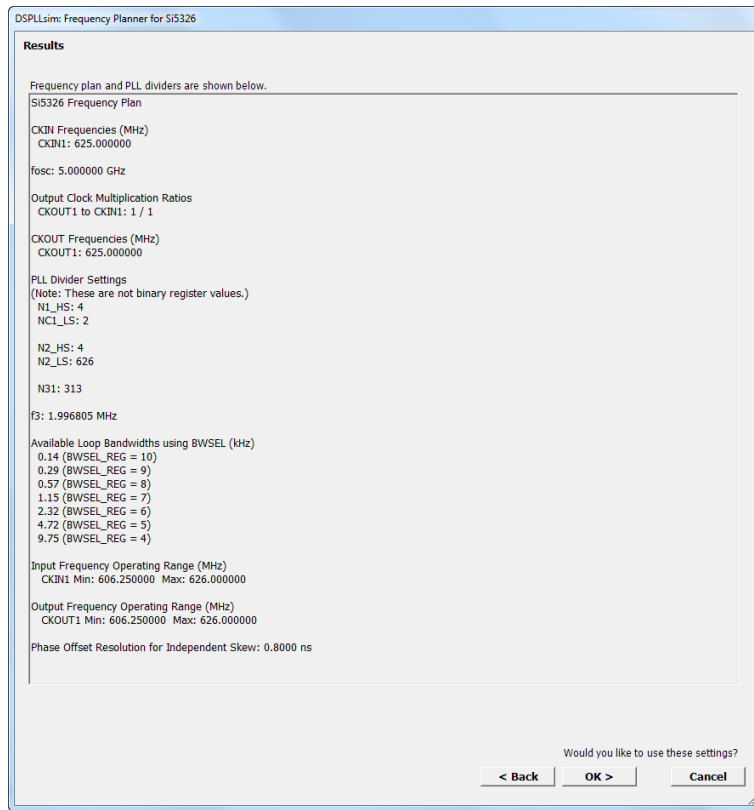


Figure 15: Frequency setting results.

After pressing OK the setup window shown in Figure 16 appears. In this window the user can select the PLL bandwidth which has an impact on the shape of the phase noise as shown in Figure 16. The user can also select the type of electrical output (CML, LVPECL, CMOS or LVDS) as shown in Figure 17.

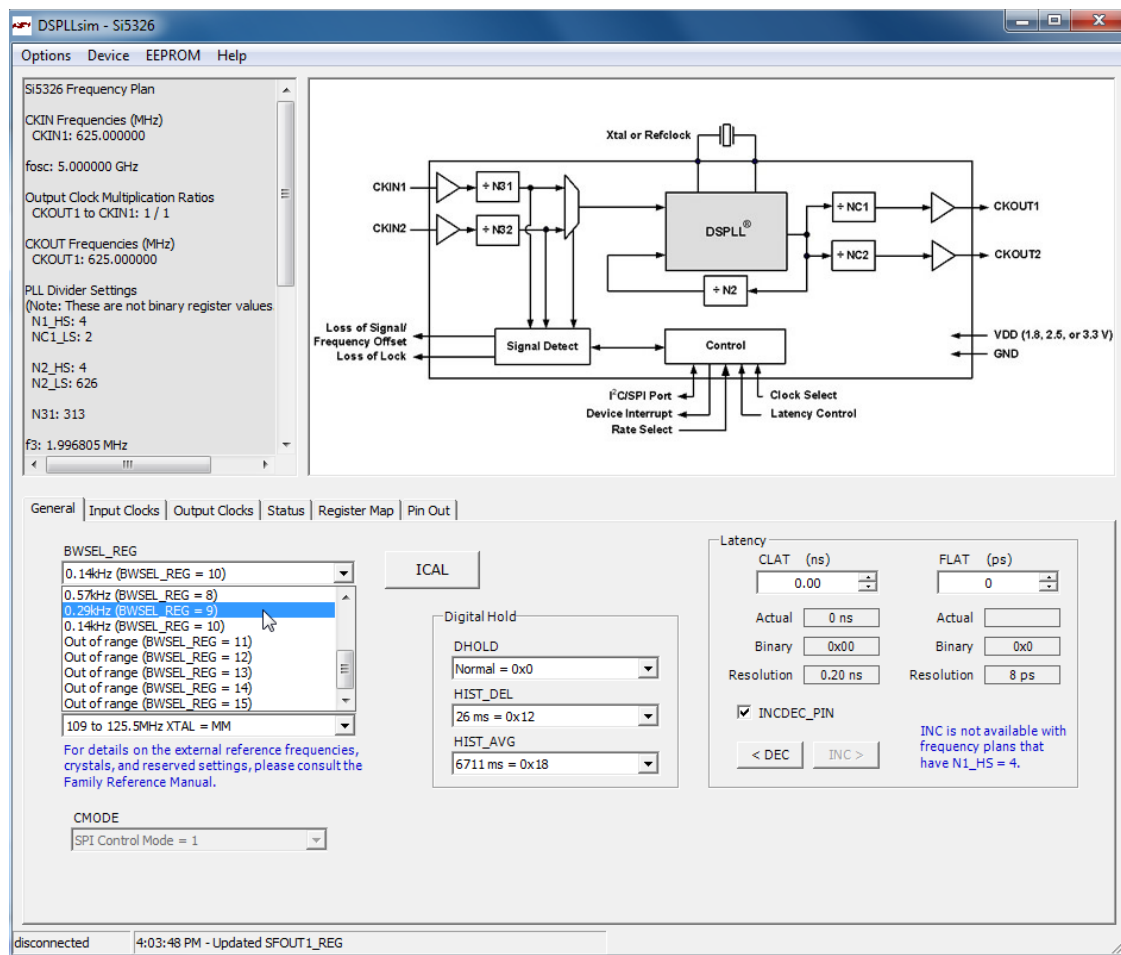


Figure 16: Selecting the PLL bandwidth.

Finally after finishing all these configuration steps it is possible to save the configuration settings to a file as shown in Figure 18.

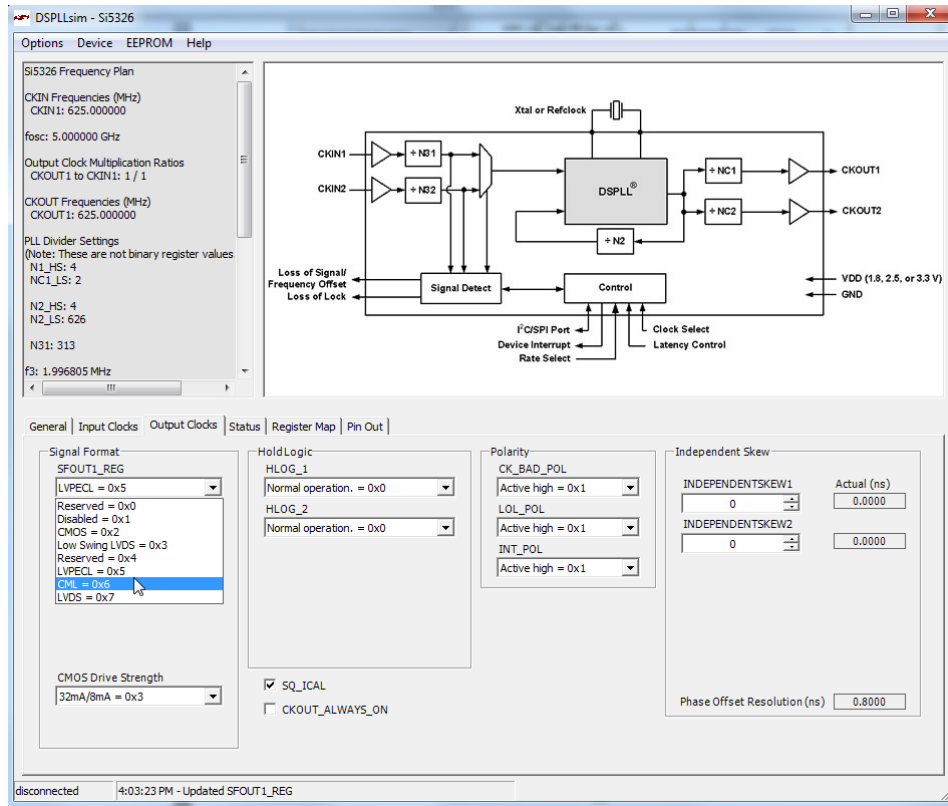


Figure 17: Selecting the output signal electrical amplitude specifications.

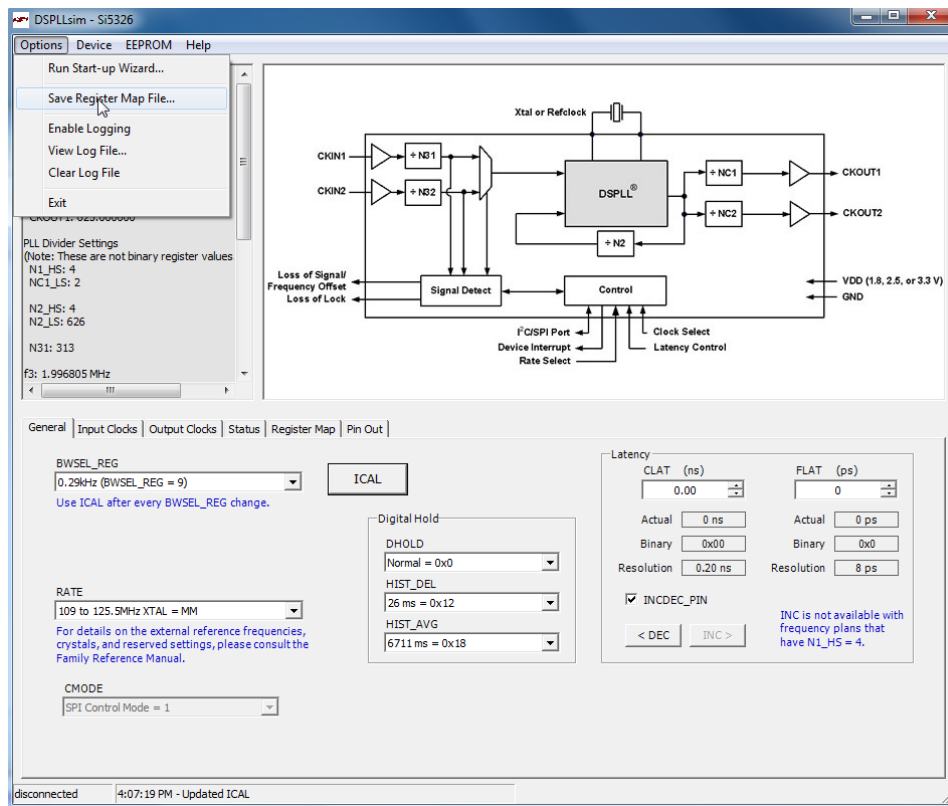


Figure 18: Saving the register map file with the jitter attenuator configuration.

The saved ASCII file with the register map will be similar to the example below. Note that the example only shows part of the complete file contents.

```
#HEADER
# Date: Tuesday, April 07, 2015 1:47 PM
# File Version: 3
# Software Name: Precision Clock EVB Software
# Software Version: 5.1
# Software Date: July 23, 2014
# Part number: Si5326
#END HEADER
#PROFILE
# Name: Si5326
#INPUT
# Name: CKIN
# Channel: 1
# Frequency (MHz): 625.000000
# N3: 313
# Maximum (MHz): 626.000000
# Minimum (MHz): 606.250000
#END_INPUT
#PLL
# Name: PLL
# Frequency (MHz): 5000.000000
# f3 (MHz): 1.996805
# N1_HS: 4
# N2_HS: 4
# N2_LS: 626
# Phase Offset Resolution (ns): 0.80000
# BWSEL_REG Option: Frequency (Hz)
# 10: 143
# 9: 286
#END_PLL
#OUTPUT
# Name: CKOUT
# Channel: 1
# Frequency (MHz): 625.000000
# NCL_LS: 2
# CKOUT1 to CKIN1 Ratio: 1 / 1
# Maximum (MHz): 626.000000
# Minimum (MHz): 606.250000
#END_OUTPUT
#CONTROL_FIELD
# Register-based Controls
#     FREE_RUN_EN: 0x0
#     CKOUT_ALWAYS_ON: 0x0
#     BYPASS_REG: 0x0
#     BWSEL_REG: 0xA
#     AUTOSEL_REG: 0x2
#     HIST_DEL: 0x12
#     ICMOS: 0x3
#     SFOUT2_REG: 0x5
#     SFOUT1_REG: 0x5
#     FOSREFSEL: 0x2
#     HLOG_2: 0x0
#     HLOG_1: 0x0
#     HIST_AVG: 0x18
#     DSBL2_REG: 0x1
#     LOS1_EN: 0x3
#     LOS2_EN: 0x3
#     FOS1_EN: 0x1
#     FOS2_EN: 0x1
#     INDEPENDENTSKEW1: 0x0
#     INDEPENDENTSKEW2: 0x0
#END_CONTROL_FIELD
#REGISTER_MAP
0, 14h
1, E4h
2, A2h
3, 15h
4, 92h
5, EDh
6, 2Dh
7, 2Ah
8, 00h
9, C0h
10, 08h
11, 42h
16, 00h
17, 80h
18, 00h
19, 2Ch
20, 3Eh
21, FFh
22, DFh
23, 1Fh
24, 3Fh
25, 00h
31, 00h
#END_REGISTER_MAP
#END_PROFILE
```

After saving the register map file we are now ready to use it on the V93000 SmarTest device setup. First the device setup needs to include all the required pins of the jitter attenuator module (SPI interface and reference clock inputs) as shown in Figure 19 for the case of a single clock reference input. Also it is necessary to define the protocol for the SPI interface as shown in Figure 20.

	Name	No	Type	Mode	HW	Connection Set	Comment	Tester Channel
								Site 1
1	CKIN1p		i	std	PS1600			12111
2	CKIN1n		i	std	PS1600			12109
3	SCL		i	std	PS1600			12103
4	SDI		i	std	PS1600			12106
5	SDO		o	std	PS1600			12104
6	A2_SS		i	std	PS1600			12101
7	RSTn		i	std	PS1600			12107

Figure 19: Pin configuration for the jitter attenuator module.

Overview

Edit the protocol structure.

type filter text

- ▼ SPI
 - idle (cycles)
 - read (slaveAddress,slaveData)
 - reset (cycles)
 - write (slaveInstruction,slaveAddressData)

Protocol Details

Set the properties of the selected protocol.

▼ Pin Roles (5 items)

A pin role is an abstract definition of a protocol pin. It is associated with a device pin in the protocol assignment. (changes affect all transactions)

Properties:

Name	Value
SSB	
MOSI	
SCLK	
MISO	
nRST	

▼ Pin Role Groups (0 items)

▼ Symbol Lists (0 items)

Define lists containing symbols.

Name	Value

▼ Constants (5 items)

Figure 20: Protocol definition of the SPI interface for the jitter attenuator module.

The jitter attenuator file can then be loaded and the jitter attenuator module programmed by using a custom designed testmethod as shown in Figure 21. The user only needs to provide the location of the register map file to be used.

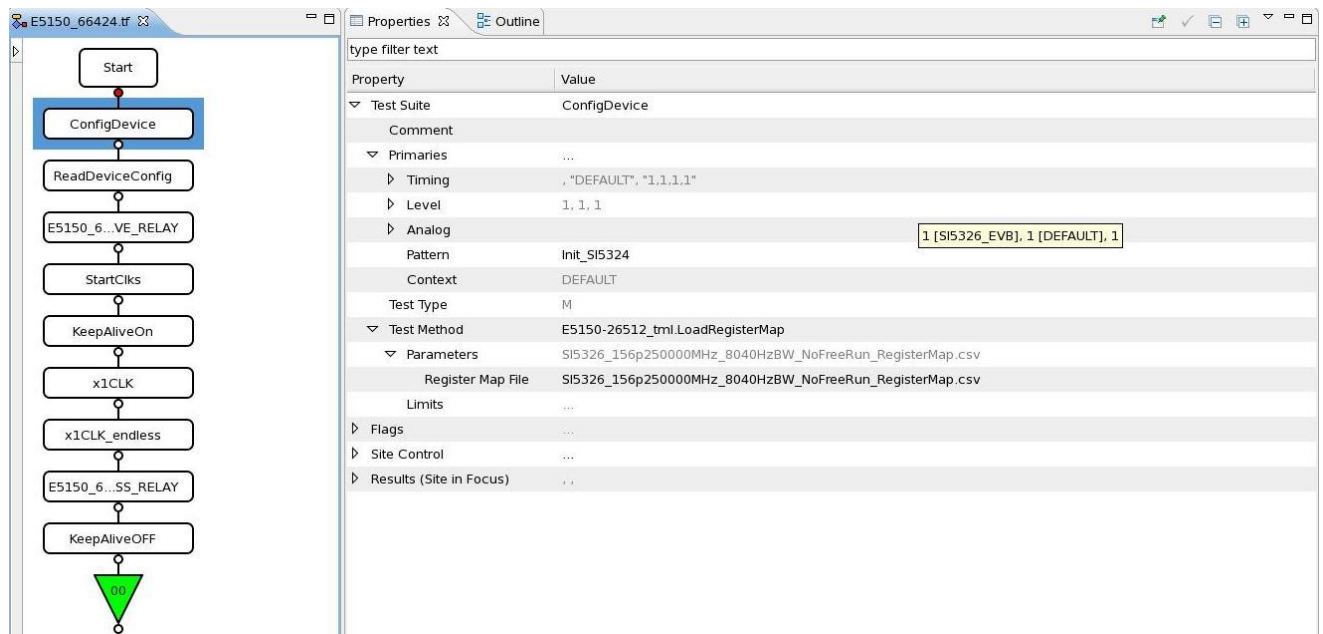


Figure 21: Testmethod to configure the jitter attenuator module.

The authors have developed a device setup including the required testmethod library to be used with the setup shown in Figure 8 that can be used as a basis for integrating the jitter attenuator module V93000 SmartTest configuration requirements on the target device configuration setup.

5. Performance Results

To measure and compare the performance of the jitter attenuator, a Keysight Technologies E5052B Signal Source Analyzer (SSA) was used since it provides the best phase noise measurement accuracy possible. Since most DUTs' reference clocks are differential and the E5052B input is single-ended, it is necessary to use a balun to measure a differential signal. We used a Marki Microwave BALH-0006 phase matched balun. Figure 8 shows a picture of the used measurement setup.

Figure 22 shows a comparison of the measured phase noise for a 156.25 MHz clock signal generated with V93000 PortScale Low Phase Noise (LPN) source, PinScale 1600 channel (PS1600), PinScale 1600 with the jitter attenuator module, PinScale 9G (PS9G), PinScale SL (PSSL) and a Rohde and Schwarz SMA100A RF source with the low phase noise option. The jitter attenuation IC (SI5326) was programmed with a PLL bandwidth of 8 KHz.

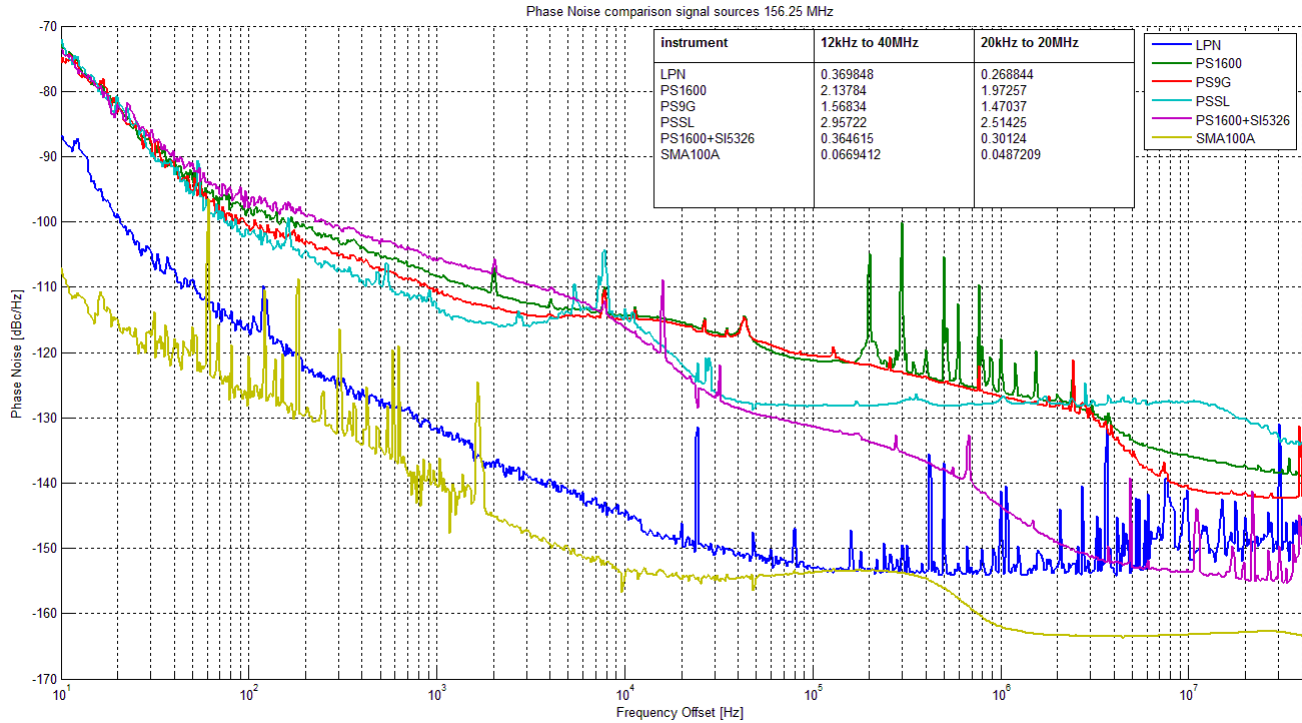


Figure 22: Measure phase noise comparison for a 156.25 MHz bit clock between the V93000 LPN, PS1600, PS9G, PSSL, PS1600 plus the jitter attenuator and the Rohde and Schwarz SMA100A RF source.

The results show as expected that the Port Scale LPN source has the best phase noise performance of all the V93000 resources but one needs to remember that is limited to a maximum frequency of 200 MHz. Although the PS1600 channels have overall the worst phase noise performance of all the V93000 digital channels presented, when used together with the jitter attenuator module it achieves a much better phase noise performance than any of the digital channels although not as good as the LPN source.

One can also observe the improvement that the jitter attenuator provides in the time domain using a real time sampling oscilloscope. For this we used a Tektronix MS073304DX real time oscilloscope. It is important to understand that correlating phase noise measurement with time domain measurements is not trivial [3] because an oscilloscope will measure all the jitter contained in the signal being measured while a phase noise measurement only measures a certain frequency band around the signal carrier. Figure 23 shows the measured jitter (time interval error or TIE) histogram for a 156.25 MHz clock signal generated from a PinScale 1600 channel. Figure 24 shows the same measurement but with the jitter attenuator connected to the PinScale 1600 channel. The jitter histogram standard deviation which corresponds to the RMS jitter if only random jitter is present, reduces from 5.4 ps to 2.4 ps

Figure 25 shows the measured phase noise from a PS1600 channel with the jitter attenuator for different target DUT reference clock frequencies. The phase noise behavior from the jitter attenuator depends on the PLL parameters especially the PLL bandwidth as shown in Figure 26 for a 156.25 MHz reference clock frequency. There are different tradeoffs as is possible to observe in Figure 26 between the phase noise behavior on the low and high frequencies and also it has an impact on the lock time of the PLL.

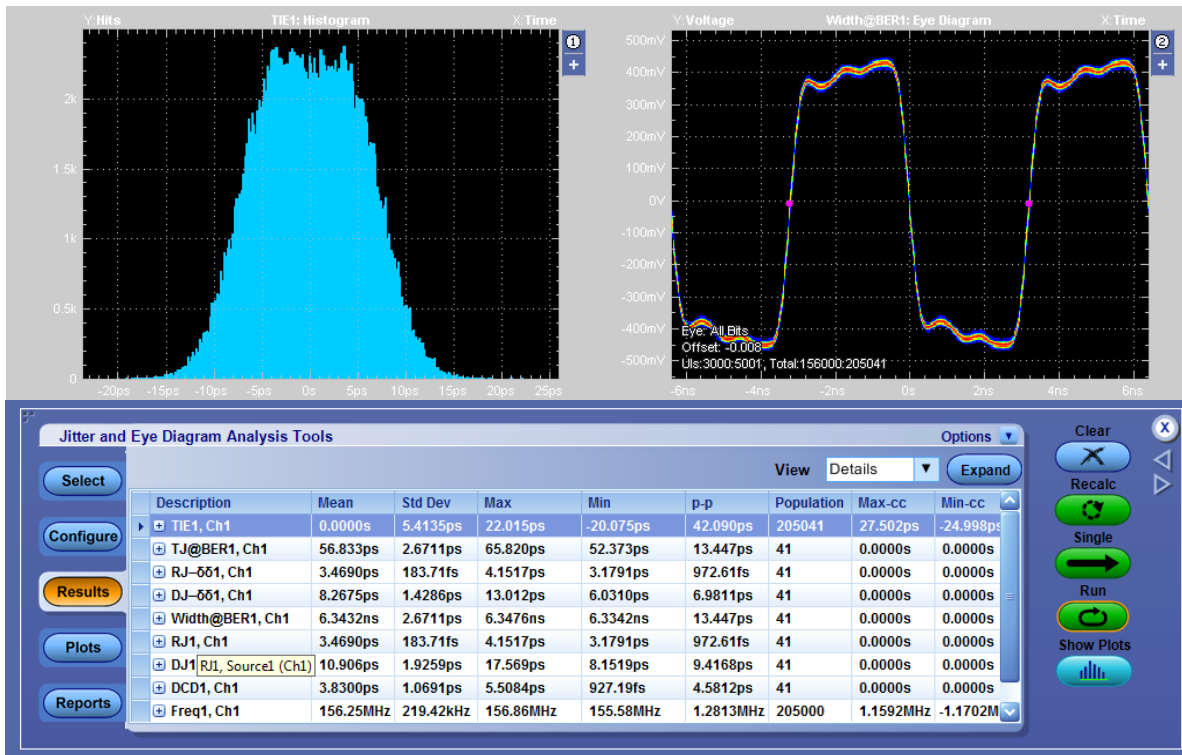


Figure 23: Measured jitter histogram from a PS1600 channel for a 156.25 MHz bit clock.

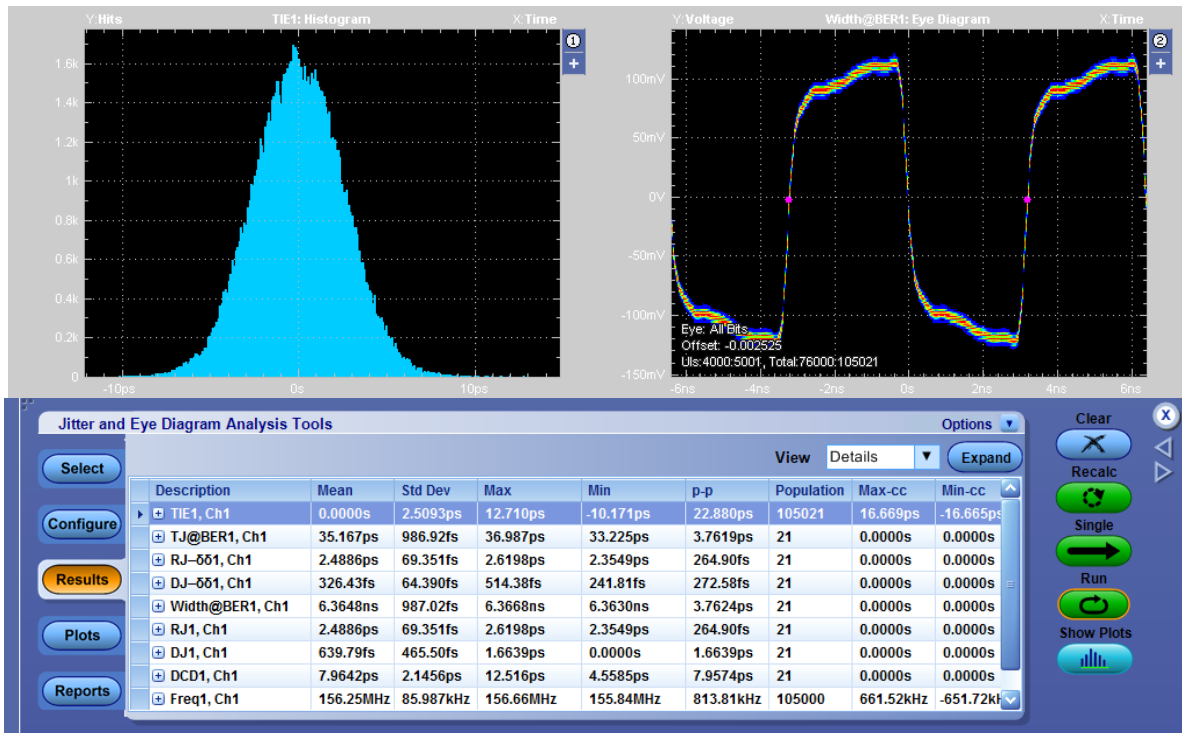


Figure 24: Measured jitter histogram from a PS1600 channel plus the jitter attenuator module for a 156.25 MHz bit clock.

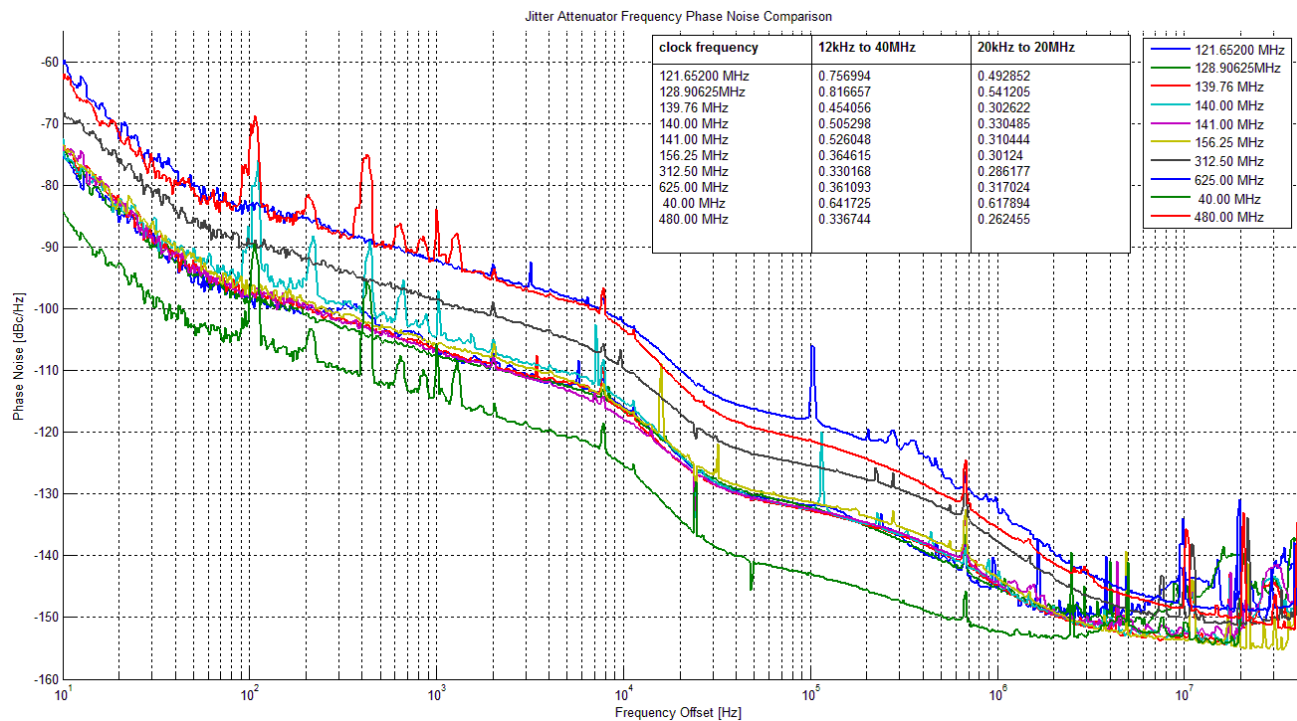


Figure 25: Jitter attenuator with PS1600 phase noise performance for different reference clock frequencies.

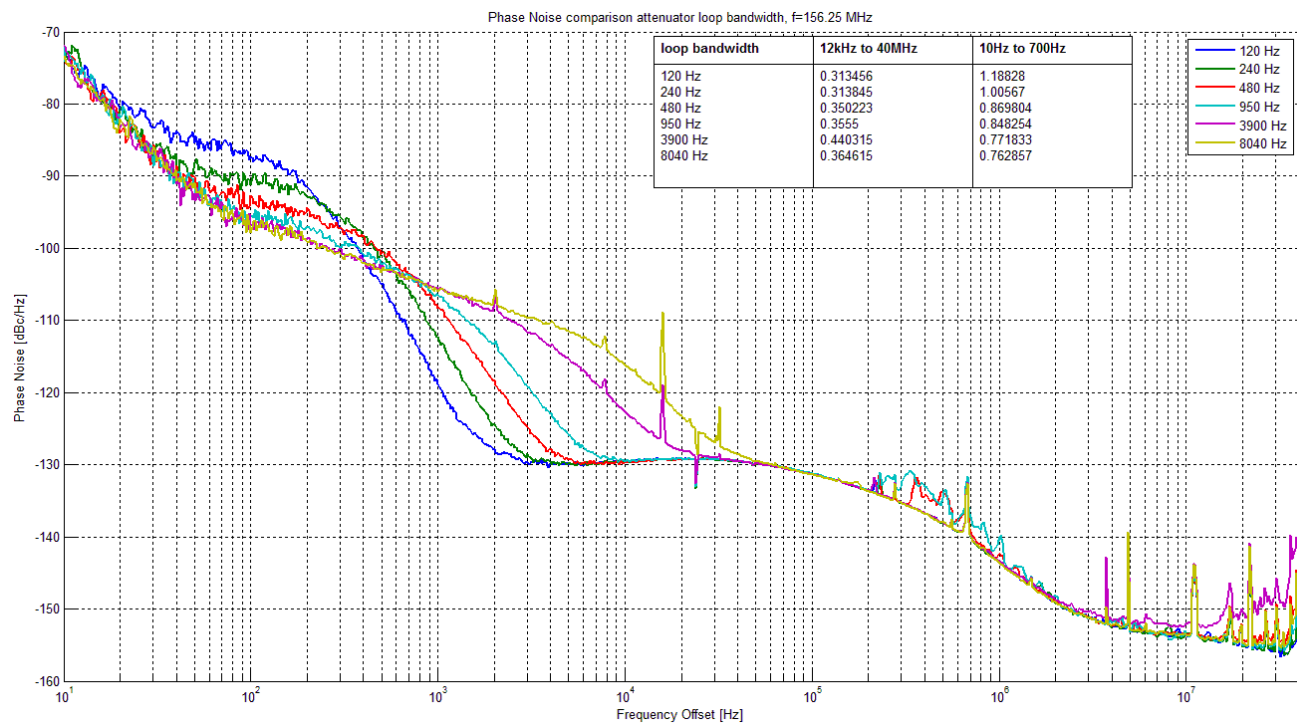


Figure 26: Jitter attenuator PLL bandwidth impact on the jitter attenuator phase noise for a 156.25 MHz reference clock (using PinScale 1600 as input reference to the jitter attenuator).

The previous results have only shown measurements taken at the output of the jitter attenuator but it is important to understand that the DUT reference clock performance has a direct impact on the DUT I/O performance as demonstrated in Figure 27 which shows a HSIO DUT transmitter of a device with PS1600 as reference clock source versus PS1600 plus jitter attenuator module as DUT reference clock source.

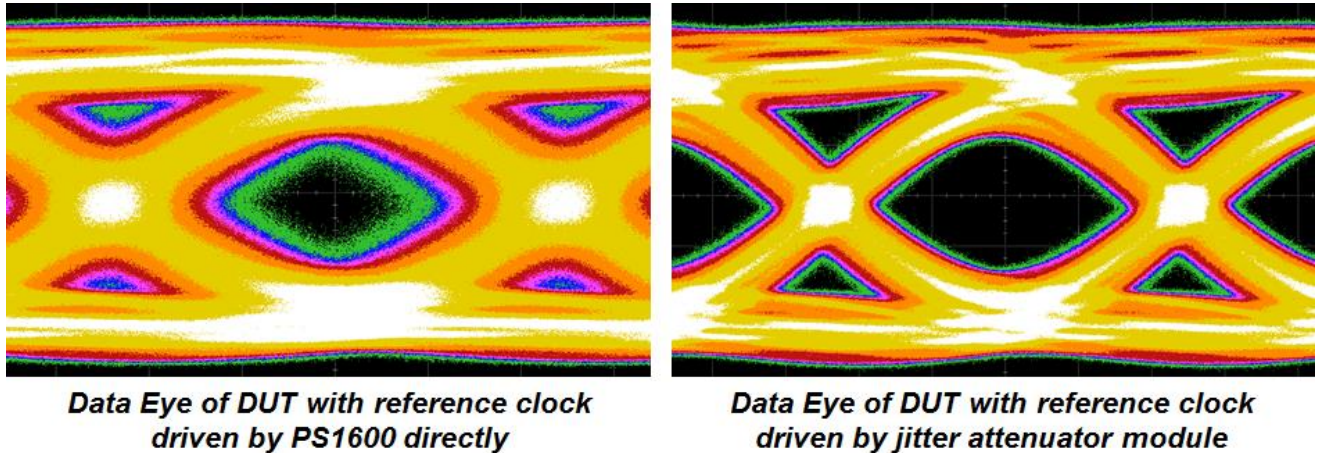


Figure 27: Impact of the DUT reference clock jitter performance on the DUT I/O performance.

6. Conclusions

This paper has presented a jitter attenuator module that together with a low cost ATE channel (e.g. PS1600) can provide a very low phase noise DUT reference clock. The module can be easily attached to the bottom of the DUT loadboard and provides a bypass path for DC tests. The main drawbacks of this approach are the additional space needed on the loadboard and the extra digital channels for the jitter attenuation module. There is also an additional lock time needed for the jitter attenuator PLL which will depend on the PLL parameters but the choice of the Si5326 IC provides a good compromise between lock time and performance. Also on a typical test program this lock time can be hidden behind other tests (e.g. DC).

7. Acknowledgments

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8. Biographies

Hubert Werkmann is currently employed as Principal Consultant at Advantest Europe GmbH, where he works on ATE test solution development for high-speed I/O interfaces and the deployment of these test solutions to leading semiconductor manufacturers. He looks back to 23 years of experience in the semiconductor design and test industry with various positions at IMS Chips, Agilent Technologies, Verigy and Advantest. Dr. Werkmann has multiple published papers in the area of scan-based design for testability and ATE high-speed testing of computation and memory devices. He received a Diploma degree in Computer Science from the University Stuttgart, Germany and holds a Ph.D. in Engineering from the University Stuttgart.

Jose Moreira is a senior staff engineer in the test cell innovation team of the SOC business unit at Advantest Boeblingen, Germany. He has a master of science in electrical and computer

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Sui-Xia Yang is an expert test engineer in application development center of the SOC business unit at Advantest Shanghai, China. She has a master degree from Beijing University of Posts and Telecommunications. She has 10 years of experience in semiconductor testing, mainly on High speed digital and mixed signal applications

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